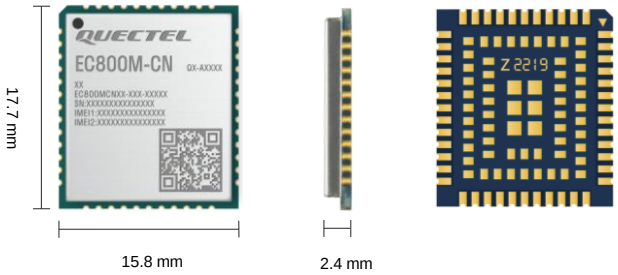
Quectel EC800M-CN

IoT/M2M-optimized LTE Cat 1 Module



EC800M-CN is a cost-effective LTE Cat 1 wireless communication module specially designed by Quectel for M2M and IoT fields. It delivers a maximum downlink rate of 10 Mbps and a maximum uplink rate of 5 Mbps. Designed in the compact and unified form factor, EC800M-CN is compatible with LTE Standard EC800E-CN, EC800G-CN and EC800N-CN in package.

EC800M-CN adopts the laser engraving process to get a more fashionable appearance, strong metallic texture, better heat dissipation, durable label information, which makes it more suitable for automation requirements.

A rich set of Internet protocols, industry-standard interfaces and abundant functionalities (USB serial drivers for Windows7/8/8.1/10/11, Linux, Android) extend the applicability of the module to a wide range of M2M and IoT applications such as OTT, Tracker, POS, Data Card, Security and Industrial PDA.

Key Features

- Ultra-small size, designed for M2M and IoT applications, especially for small-size terminals
- Support remote DFOTA, reducing later operation and maintenance costs
- Support Wi-Fi Scan
- Super cost-effective
- Dual SIM Dual Standby Single Pass (Optional)

LTE Cat 1
Max. 10 Mbps (DL)
Max. 5 Mbps (UL)

LCC Package

Embedded
Abundant Protocols

USB 2.0 High Speed Interface

USB Drivers

VoLTE (Optional)

Quectel Enhanced AT Commands

DFOTA

Dual SIM Dual Standby Single Pass

Version: 1.0 | Status: Released

Quectel EC800M-CN

LTE Cat 1		EC800M-CN
Region/Operator		China/India
Package		LCC
Dimensions		17.7 mm × 15.8 mm × 2.4 mm
Weight		Approx. 1.4 g
Temperature Range		
Operating Temperature Range		-35 °C to +75 °C
Extended Temperature Range		-40 °C to +85 °C
Frequency Bands		
LTE-FDD		B1/3/5/8
LTE-TDD		B34/38/39/40/41
GNSS (Optional)		GPS/BDS/GLONASS/Galileo
Certifications		
Regulatory		**China:** SRRC*/NAL*/CCC*
Others		WHQL
Data Rates		
LTE-FDD (Mbps)		10 (DL)/5 (UL)
LTE-TDD (Mbps)		8.96 (DL)/3.1 (UL)
Interfaces		
USIM①		× 2, 1.8/3.0 V
UART		× 3 (Main, Debug and Auxiliary*)
USB 2.0		× 1
Analog Audio Input (× 1) (Optional) **Analog Audio Output (× 1)**		× 1 (Microphone/Earpiece)
PCM (Digital Audio)		× 1
ADC		× 2
NET_STATUS		× 1
STATUS		× 1
I2C		× 2
LCM②		× 1
Camera②		× 1
Matrix Keyboard (5 × 5)②		× 1
USB_BOOT		× 1
RESET_N		× 1
PWRKEY		× 1
Antennas		× 2 (Main Antenna and GNSS Antenna (Optional))
Voice		
Speech Codec Modes		MP3/AMR/WAV
Voice		Digital Audio, Analog Audio and VoLTE (Voice over LTE)
Enhanced Features		
DFOTA		Supported
DTMF		Optional
Audio Playback*/Audio Recording*		Optional
QuecOpen®		Optional
QuecLocator®		Optional
QuecFile®		Supported
(U)SIM1 Card Detection		Supported
Software Features		
Protocols		TCP/UDP/PPP/NTP/NITZ/FTP/HTTP/PING/CMUX*/HTTPS/FTPS/SSL/FILE/MQTT/MMS*/SMTP*/SMTPS*
Drivers	**RIL**	Android 4.x~12.x
	USB RNDIS	Windows 7/8/8.1/10/11 Linux 2.6~5.18
	USB ECM	Linux 2.6~5.18
	USB Serial	Windows 7/8/8.1/10/11 Linux 2.6~5.18 Android 4.x~12.x
Electrical Features		
Supply Voltage Range		3.4~4.3 V， typ. 3.8 V
Power Consumption (Typical)		39 μA @ Power off 1.0 mA@ LTE Sleep (PF = 128) 0.9 mA @ LTE Sleep (PF = 256) 16.5 mA @ Idle (PF = 64 USB disconnected) 29.1 mA @ Idle (PF = 64 USB connected)

NOTE:
1. *: under development/in progress.
2. ①: USIM2 is optional.
3. ②: Supported only by the QuecOpen® solution.

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